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Roll No

EC-702(C)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Nano Electronics

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain the different types of PVD techniques.
b) Explain the theory involved in integer quantum hall effect.
2. a) What is the major difference between proximity printing and projection printing. Write down the limitation of proximity printing.
b) Write short note on : Electron lithography and Electron resists.
3. a) Illustrate the principle of operation of Resonant tunneling diode.
b) Explain the concept of coulomb blockade. Derive the conditions to be fulfilled to observe single electron effect?
4. a) With the help of neat schematic diagram Explain MOSFEETS.
b) Explain the principle of operation of a quantum dot based single electron transistor.

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5. a) Write short note on NEMS.
b) Explain the working of a chemical sensor.
6. a) Compare and contrast the features of square, triangular and parabolic quantum wells.
b) Describe Super lattice.
7. a) Derive expression for the voltage required for the tunneling through a nano capacitor.
b) Explain the structure of MODFET.
8. a) Explain the principle of Scanning Tunneling Microscope with neat diagrams.
b) Explain parabolic and triangular quantum wells with neat diagrams.
